

Important information. Please read carefully.

ALOVASC

Tablet 5 mg / 10 mg

COMPOSITION

Alovasc 5 mg: Each tablet contains Amlodipine Besylate equivalent to Amlodipine 5 mg

Alovasc 10 mg: Each tablet contains Amlodipine Besylate equivalent to Amlodipine 10 mg

PHARMACODYNAMICS

Amlodipine is a calcium ion influx inhibitor of the dihydropyridine group (slow channel blocker or calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into cardiac and vascular smooth muscle.

The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on vascular smooth muscle. The precise mechanism by which amlodipine relieves angina has not been fully determined but amlodipine reduces total ischaemic burden by the following two actions:

1. Amlodipine dilates peripheral arterioles and thus, reduces the total peripheral resistance (afterload) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.
2. The mechanism of action of amlodipine also probably involves dilatation of the main coronary arteries and coronary arterioles, both in normal and ischaemic regions. This dilatation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina).

In patients with hypertension, once daily dosing provides clinically significant reductions of blood pressure in both the supine and standing positions throughout the 24 hour interval. Due to the slow onset of action, acute hypotension is not a feature of amlodipine administration.

In patients with angina, once daily administration of amlodipine increases total exercise time, time to angina onset, and time to 1mm ST segment depression, and decreases both angina attack frequency and nitroglycerine tablet consumption.

Amlodipine has not been associated with any adverse metabolic effects or changes in plasma lipids and is suitable for use in patients with asthma, diabetes, and gout.

PHARMACOKINETICS

Absorption, distribution, plasma protein binding: After oral administration of therapeutic doses, amlodipine is well absorbed with peak blood levels between 6-12 hours post dose. The absorption of amlodipine is unaffected by the concomitant intake of food. Absolute bioavailability has been estimated to be between 64 and 80%. The volume of distribution is approximately 21 L/kg. *In vitro* studies have shown that approximately 97.5% of circulating amlodipine is bound to plasma proteins.

Biotransformation/elimination: The terminal plasma elimination half life is about 35-50 hours and is consistent with once daily dosing. Steady state plasma levels are reached after 7-8 days of consecutive dosing. Amlodipine is extensively metabolised by the liver to inactive metabolites with 10% of the parent compound and 60% of metabolites excreted in the urine.

Use in the elderly: The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly patients. Increases in AUC and elimination half-life in patients with congestive heart failure were as expected for the patient age group studied.

INDICATIONS

Hypertension: Alovasc is indicated for the first line treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of Alovasc, which has been used in combination with a thiazide diuretic, alpha blockers, beta adrenoceptor blocking agent, or an angiotensin converting enzyme inhibitor.

Myocardial Ischemia: Alovasc is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Alovasc may be used where the clinical presentation suggests a possible vasospastic/vasoconstrictive component but vasospastic/vasoconstrictive has not been confirmed. Alovasc may be used alone, as monotherapy, or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of beta-blockers.

DOSAGE AND ADMINISTRATION

Adults: For both hypertension and angina the usual initial dose is 5 mg Alovasc once daily which may be increased to a maximum dose of 10 mg depending on the individual patient's response. No dose adjustment of Alovasc is required upon concomitant administration of thiazide diuretics, beta-blockers and angiotensin-converting enzyme inhibitors.

In children: Not recommended.

In the elderly: Normal dosage regimens are recommended. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated.

Patients with hepatic impairment: See section Precautions.

Patients with renal impairment: Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialyzable.

CONTRAINDICATIONS

Amlodipine is contra-indicated in patients with a known sensitivity to dihydropyridines, amlodipine or any of the excipients.

Amlodipine should not be used in cardiogenic shock, clinically significant aortic stenosis, unstable angina (excluding Prinzmetal's angina).

WARNINGS AND PRECAUTIONS

The safety and efficacy of amlodipine in hypertensive crisis has not been established.

Patients with cardiac failure

Patients with heart failure should be treated with caution. Calcium channel blockers, including amlodipine, should be used with caution in patients with congestive heart failure, as they may increase the risk of future cardiovascular events and mortality.

Patients with hepatic impairment

The half-life of amlodipine is prolonged and AUC values are higher in patients with impaired liver function; dosage recommendations have not been established. Amlodipine should therefore be initiated at the lower end of the dosing range and caution should be used, both on initial treatment and when increasing the dose. Slow dose titration and careful monitoring may be required in patients with severe hepatic impairment.

Elderly patients

In the elderly increase of the dosage should take place with care.

Patients with renal impairment

Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialysable.

DRUG INTERACTIONS

Effects of other medicinal products on amlodipine

CYP3A4 inhibitors

Concomitant use of amlodipine with strong or moderate CYP3A4 inhibitors (protease inhibitors, azole antifungals, macrolides like erythromycin or clarithromycin, verapamil or diltiazem) may give rise to significant increase in amlodipine exposure resulting in an increased risk of hypotension. The clinical translation of these pharmacokinetics variations may be more pronounced in the elderly. Clinical monitoring and dose adjustment may thus be required.

CYP3A4 inducers

Upon co-administration of known inducers of the CYP3A4, the plasma concentration of amlodipine may vary. Therefore, blood pressure should be monitored and dose regulation considered both during and after concomitant medication particularly with strong CYP3A4 inducers (e.g. rifampicin, hypericum perforatum).

Administration of amlodipine with grapefruit or grapefruit juice is not recommended as bioavailability may be increased in some patients resulting in increased blood pressure lowering effects.

Dantrolene (infusion)

In animals, lethal ventricular fibrillation and cardiovascular collapse are observed in association with hyperkalemia after administration of verapamil and intravenous dantrolene. Due to risk of hyperkalemia, it is recommended that the co-administration of calcium channel blockers such as amlodipine be avoided in patients susceptible to malignant hyperthermia and in the management of malignant hyperthermia.

Effects of amlodipine on other medicinal products

The blood pressure lowering effects of amlodipine adds to the blood pressure-lowering effects of other medicinal products with antihypertensive properties.

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Client : Good Science Brand : Alovasc Tablets 5mg/10mg Product Code : PM00140/07-08-23/REV01 Barcode : NA Date : 24 / 08 / 23	INTERNAL Ref. No. : WS-GSK-23-185 Folder Name : GS Leaflet Alovasc PM00140-070823-R1 File Name : PM00140/07-08-23/REV01 Alovasc Leaflet FRONT_ok Designer : Aini Check By : WSam	PLEASE TAKE NOTE ! While every care is taken to meet client’s requirements, it is the client’s responsibility to check and ensure all information and specifications on this artwork are correct before proceeding to the next stage of production. MUN SUN cannot accept responsibility for any errors or omissions.	CLIENTS APPROVAL Approved By / Name: _____ Date: _____ _____ _____ _____	COMMENTS 24 / 08 / 23 rev.1 1. Used existing AW to amend. 2. Bold word “Absorption, distribution, plasma…….” 29 / 09 / 23 rev.2 1. Added item code PM00140/27-09-23/REV002 11 / 10 / 23 rev.3 1. Revised artwork. Added text. 23 / 10 / 23 rev.4 1. Beta blockers to beta-blockers
COLOURS TO USE: K				

Tacrolimus

There is a risk of increased tacrolimus blood levels when co-administered with amlodipine but the pharmacokinetic mechanism of this interaction is not fully understood. In order to avoid toxicity of tacrolimus, administration of amlodipine in a patient treated with tacrolimus requires monitoring of tacrolimus blood levels and dose adjustment of tacrolimus when appropriate.

Mechanistic Target of Rapamycin (mTOR) Inhibitors

mTOR inhibitors such as sirolimus, temsirolimus, and everolimus are CYP3A substrates. Amlodipine is a weak CYP3A inhibitor. With concomitant use of mTOR inhibitors, amlodipine may increase exposure of mTOR inhibitors.

Cyclosporine

No drug interaction studies have been conducted with cyclosporine and amlodipine in healthy volunteers or other populations with the exception of renal transplant patients, where variable trough concentration increases (average 0% - 40%) of cyclosporine were observed. Consideration should be given for monitoring cyclosporine levels in renal transplant patients on amlodipine, and cyclosporine dose reductions should be made as necessary.

Simvastatin

Co-administration of multiple doses of 10 mg of amlodipine with 80 mg simvastatin resulted in a 77% increase in exposure to simvastatin compared to simvastatin alone. Limit the dose of simvastatin in patients on amlodipine to 20 mg daily.

In clinical interaction studies, amlodipine did not affect the pharmacokinetics of atorvastatin, digoxin or warfarin.

PREGNANCY AND LACTATION

Safety of amlodipine in human pregnancy or lactation has not been established. Amlodipine does not demonstrate toxicity in animal reproductive studies other than to delay parturition and prolong labor in rats at dose level fifty times the maximum recommended dose in humans. Accordingly, use in pregnancy is only recommended when there is no safer alternative and when the disease itself carries greater risk for the mother and fetus.

SIDE EFFECTS / ADVERSE REACTIONS

The most commonly reported adverse reactions during treatment are somnolence, dizziness, headache, palpitations, flushing, abdominal pain, nausea, ankle swelling, oedema and fatigue. Exceptional cases of extrapyramidal syndrome have been reported.

Blood and lymphatic system disorders

Very rare: Leukocytopenia, thrombocytopenia

Immune system disorders

Very rare: Allergic reactions

Metabolism and nutrition disorders

Very rare: Hyperglycaemia

Psychiatric disorders

Uncommon: Depression, mood changes (including anxiety), insomnia

Rare: Confusion

Nervous system disorders

Common: Somnolence, dizziness, headache (especially at the beginning of the treatment)

Uncommon: Tremor, dysgeusia, syncope, hypoaesthesia, paraesthesia

Very rare: Hypertonia, peripheral neuropathy

Eye disorders

Common: Visual disturbance (including diplopia)

Ear and labyrinth disorders

Uncommon: Tinnitus

Cardiac disorders

Common: Palpitations

Uncommon: Arrhythmia (including bradycardia, ventricular tachycardia and atrial fibrillation)

Very rare: Myocardial infarction

Vascular disorders

Common: Flushing

Uncommon: Hypotension

Very rare: Vasculitis

Respiratory, thoracic and mediastinal disorders

Common: Dyspnoea

Uncommon: Cough, rhinitis

Gastrointestinal disorders

Common: Abdominal pain, nausea, dyspepsia, altered bowel habits (including diarrhoea and constipation)

Uncommon: Vomiting, dry mouth

Very rare: Pancreatitis, gastritis, gingival hyperplasia

Hepatobiliary disorders

Very rare: Hepatitis, jaundice, hepatic enzyme increased

Skin and subcutaneous tissue disorders

Uncommon: Alopecia, purpura, skin discolouration, hyperhidrosis, pruritus, rash, exanthema, urticarial

Very rare: Angioedema, erythema multiforme, exfoliative dermatitis, Stevens-Johnson syndrome, Quincke oedema, photosensitivity

Not known: Toxic epidermal necrolysis

Musculoskeletal and connective tissue disorders

Common: Ankle swelling, muscle cramps

Uncommon: Arthralgia, myalgia, back pain

Renal and urinary disorders

Uncommon: Micturition disorder, nocturia, increased urinary frequency

Reproductive system and breast disorders

Uncommon: Impotence, gynaecomastia

General disorders and administration site conditions

Very common: Oedema

Common: Fatigue, asthenia

Uncommon: Chest pain, pain, malaise

Investigations

Uncommon: Weight increased, weight decreased

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Amlodipine can have minor or moderate influence on the ability to drive and use machines. If patients taking amlodipine suffer from dizziness, headache, fatigue or nausea the ability to react may be impaired. Caution is recommended especially at the start of treatment.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE

Available data suggest that gross overdosage could result in excessive peripheral vasodilatation and possibly reflex tachycardia. Marked and probably prolonged systemic hypotension up to and including shock with fatal outcome have been reported.

Gastric lavage may be worthwhile in some cases. Clinically significant hypotension due to amlodipine overdosage calls for active cardiovascular support including frequent monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade. Since amlodipine is highly proteinbound, dialysis is not likely to be of benefit.

STORAGE CONDITION

Store below 30°C

SHELF LIFE

The expiry date is indicated on the packaging.

PRODUCT DESCRIPTION

Alovasc 5mg:

White colour, oval shape (8,5mm x 5,0mm) normal convex tablet, engraved with "ΛV" and break line on one side and plain on the reverse. The breakline is only to facilitate breaking for ease of swallowing and not to divide into equal doses.

Available in blister pack of 10's in packaging of 30's and 100's per box.

Alovasc 10mg:

White colour, oval shape (11,0mm x 6,0mm) normal convex tablet, engraved with "ΛV" and break line on one side and plain on the reverse. The breakline is only to facilitate breaking for ease of swallowing and not to divide into equal doses.

Available in blister pack of 10's in packaging of 30's and 100's per box.

KEEP OUT OF REACH OF CHILDREN

JAUHI DARI KANAK-KANAK

For further information, please consult your pharmacist or physician.

NAME AND ADDRESS OF MANUFACTURERS

Packed and released by:

GoodScience Sdn Bhd

No. 7, Jalan KPK 4/3, Kawasan Perindustrian Kundang, Kundang Jaya, 48020 Rawang,Selangor, Malaysia.

Manufactured by:

Xepa-Soul Pattinson (Malaysia) Sdn Bhd

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PRODUCT REGISTRATION HOLDER

GoodScience Sdn Bhd

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Revision Date: 10-May-2023

PM-00140/04-10-23/REV02

240mm

INK COVERAGE		
Ink Name	mm2	%
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M	0	0
Y	0	0
K	3501.0	9.12
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Client : Good Science		INTERNAL	PLEASE TAKE NOTE !	CLIENTS APPROVAL	COMMENTS
Brand : Alovasc Tablets 5mg/10mg		Ref. No. : WS-GSK-23-185	While every care is taken to meet client's requirements, it is the client's responsibility to check and ensure all information and specifications on this artwork are correct before proceeding to the next stage of production. MUN SUN cannot accept responsibility for any errors or omissions.	Approved By / Name:	Date:
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K					24 / 08 / 23 rev.1 1. Used existing AW to amend. 2. Bold word "Absorption, distribution, plasma....." 29 / 09 / 23 rev.2 1. Added item code PM00140/27-09-23/REV002 11 / 10 / 23 rev.3 1. Revised artwork. Added text. 23 / 10 / 23 rev.4 1. Protein-bound to proteinbound 2. 6.00mm to 6.0mm 3. PM00140/07-08-23/REV01 to PM-00140/04-10-23/REV02